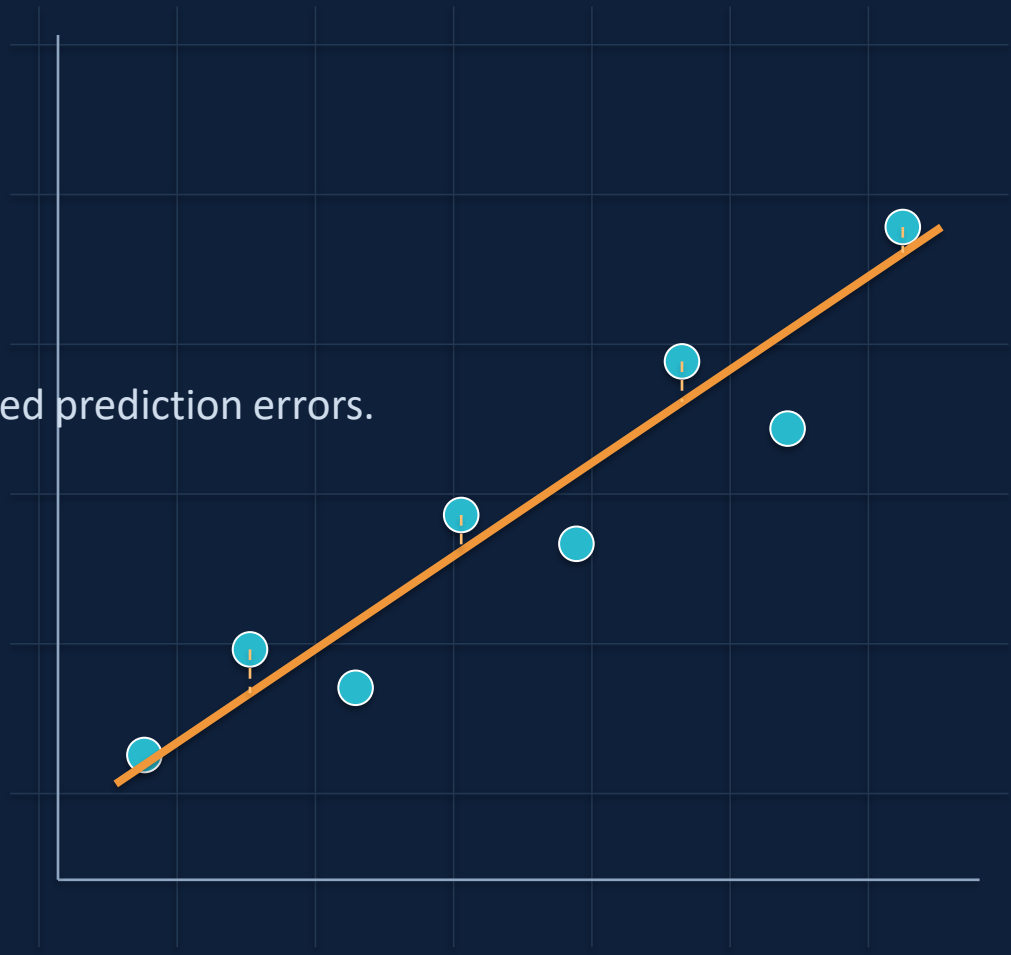
MACHINE LEARNING • LINEAR MODELS

ORDINARY LEAST SQUARES

How a linear model finds the best-fitting line by minimizing squared prediction errors.



OLS | REGRESSION

Prepared as a learning presentation

What does Linear Regression learn?

OLS estimates the coefficients of a linear regression model.

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \ldots + w_px_p$$

Prediction = intercept + weighted contribution of each feature

$x_1, x_2, \ldots$

INPUT FEATURES

Information used for prediction

$w_1, w_2, \ldots$

COEFFICIENTS

How strongly features contribute

w_0

INTERCEPT

Baseline when features are zero

NUMERICAL EXAMPLE

$$\hat{y} = 10 + 100x_1 + 80x_2$$

If $x_1 = 0.5$ and $x_2 = 0.5$:
$\hat{y} = 10 + 50 + 40 = 100$

The model predicts 100.

OLS estimates weights to make predictions fit the observed data as closely as possible.

Residuals and Sum of Squared Errors

A residual is the difference between an actual value and its predicted value.

$$\text{Residual} = y - \hat{y}$$

$$SSE = \Sigma (y_i - \hat{y}_i)^2$$

Add squared residuals across observations.

WHY SQUARE THE ERRORS?

- Positive and negative errors cannot cancel.
- Large errors are penalized more.

3-STUDENT EXAMPLE

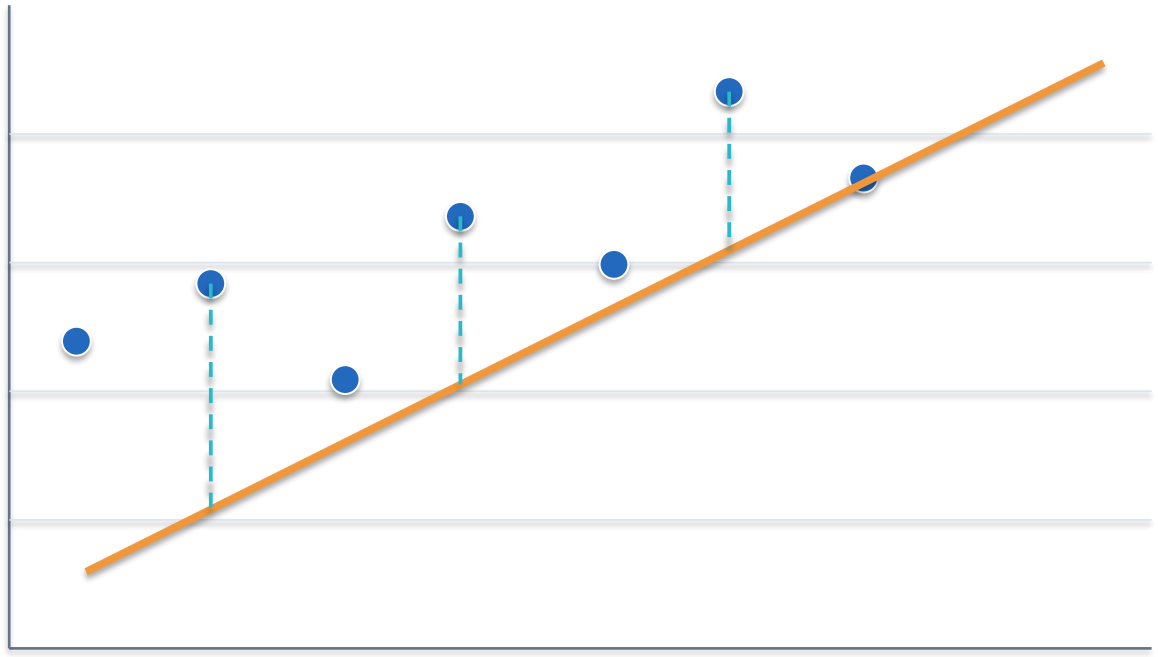
Student	Actual y	Predicted ŷ	Residual	Squared
A	80	100	−20	400
B	70	80	−10	100
C	90	90	0	0

SSE = 400 + 100 + 0 = 500

OLS objective: find weights that minimize SSE.

How does OLS choose the best-fitting line?

It chooses coefficients that make the total squared residual as small as possible.



Blue dots = observed data • Orange = fitted line • Dashed = residuals

OLS MINIMIZATION

minimize $\Sigma(y_i - \hat{y}_i)^2$

Best fit = the candidate line with the smallest SSE.

WHAT THE MODEL ESTIMATES

- Intercept (w_o)
- Feature coefficients (w_1, w_2, …)
- A line or plane that fits the training data

How can the weights be updated?

Gradient Descent is one way to move the parameters toward a lower SSE.

New weight = Old weight − η × Gradient

η (eta) is the learning rate.

ILLUSTRATIVE UPDATE

Parameter	Old value	Gradient	Updated value
w_0	10	60	9.40
w_1	100	26	99.74
w_2	80	30	79.70

EXAMPLE: η = 0.01

$w_0 = 10 - 0.01(60) = 9.4$

$w_1 = 100 - 0.01(26) = 99.74$

$w_2 = 80 - 0.01(30) = 79.70$

UPDATED MODEL

$\hat{y} = 9.4 + 99.74x_1 + 79.7x_2$

Gradient tells how SSE changes when a parameter changes.

OLS: the key ideas to remember

Ordinary Least Squares estimates coefficients by minimizing residual squared error.

MODEL

$$\hat{y} = w_0 + w_1 x_1 + \ldots$$

Weighted combination of input features.

ERROR

$$SSE = \Sigma(y_i - \hat{y}_i)^2$$

Total squared prediction error.

GOAL

minimize SSE

Choose coefficients that fit observed data.

SCIKIT-LEARN IMPLEMENTATION

```
from sklearn.linear_model import LinearRegression
```

model = LinearRegression() • model.fit(X, y)

REMEMBER

OLS is a regression method. It does not add a regularization penalty.

NEXT → Ridge Regression • Lasso Regression • Elastic Net